Amprius Technologies (AMPX)

Capital-Light Momentum Meets High-Energy Demand: Amprius Is Poised for a Multi-Year Upside

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> Amprius Technologies has transitioned from a loss-making, asset-heavy model to a capital-light contract-manufacturing engine that is now delivering double-digit revenue growth and rapidly improving gross margins. The market continues to price the stock as a speculative play on a single contract, ignoring the breadth of recent multi-year deals and the emerging margin trajectory. We believe the blended target of $5.71, representing a -24.8% upside versus the $7.59 reference price, captures the valuation gap while accounting for upside and downside scenarios.

HOLD	$5.71	-24.8%	$7.59
Hold — Valuation Concerns	BLENDED TARGET	BLENDED UPSIDE	REFERENCE PRICE
INVESTMENT RATING			

The market is fixated on Amprius's near-term cash burn, yet the company's revenue trajectory and margin expansion signal a compounding upside that will unfold over the next five years.

EXECUTIVE SUMMARY

Amprius Technologies develops silicon-anode lithium-ion batteries for drones, aviation, and electric vehicles. The business model is contract-manufacturing focused, leveraging a capital-light approach that scales with order flow rather than fixed plant capacity. This structure positions the company to capture high-energy-density demand without the balance-sheet drag of traditional battery manufacturers. Recent quarterly results demonstrate a clear inflection. Q1 FY2026 revenue surged 153% YoY to $28.5 million with gross margin flipping to 20% from a negative 21% a year earlier. Q2 FY2026 revenue doubled to $34.0 million, gross margin rose to 27%, and adjusted EBITDA narrowed to a $1.0 million loss, the closest the company has come to breakeven. Management raised FY2026 revenue guidance to at least $140 million and gross margin to at least 28%, underscoring confidence in the emerging SiCore platform. The balance sheet remains strong. Cash sits at $90.5 million against total debt of $39.9 million, yielding net cash of $50.6 million. Diluted shares total 143.5 million, giving a market capitalisation of $1.09 billion at the $7.59 reference price. The company's WACC of 11.29% reflects a beta of 1.80, indicating a risk profile above the market but justified by the nascent technology and defense exposure.

OPERATIONAL SNAPSHOT

$7.59	$71.9M	$-35.5M	-49.4%	1.80	$7.59
REF. PRICE	FY2025 REVENUE	FY2025 FREE CASH FLOW	FCF MARGIN	EQUITY BETA	REF. CLOSE

Fiscal-year figures as reported in company filings and collected by Cascadia Analytics; ref. price as of 2026-10-20.

THE MOAT IS TECHNICAL AND DEFENSIBLE

Amprius's silicon-anode technology delivers energy densities that exceed conventional graphite anodes, creating a clear performance advantage for weight-sensitive platforms such as UAVs and electric aircraft. The proprietary SiCore cell architecture is protected by a portfolio of patents that limit direct replication by competitors, raising switching costs for customers that have already qualified the cells for flight-critical applications. Regulatory barriers reinforce the moat. The company's cells meet NDAA-compliant internal component standards, a prerequisite for U.S. defense contracts. This compliance has already translated into a $14.8 million DIU contract and a $35 million follow-on order from a leading UAS manufacturer, illustrating how regulatory alignment can lock in high-value customers. Customer retention is further bolstered by the capital-light model. Amprius does not own the manufacturing lines that produce the cells; instead, it contracts with established fabs, allowing rapid scale-up without the capital constraints that plague traditional battery makers. This flexibility enables the firm to meet large, multi-year orders such as the $100 million+ Stark Future agreement while preserving cash. Pricing power is emerging as gross margins improve. The shift from a negative 21% margin in Q4 FY2025 to 27% in Q2 FY2026 reflects both learning-curve efficiencies and the ability to command premium pricing for high-energy-density solutions in defense and aerospace markets.

VALUATION

Three scenarios make the valuation range explicit

The base case assumes revenue compounds at a 38.5% CAGR through FY2030, delivering a terminal free-cash-flow (FCF) margin of 14.3% and a terminal growth rate of 3.5%. The bear case reduces the revenue CAGR to 18.0% and the terminal FCF margin to 9.3%, while the bull case accelerates revenue growth to 55.0% with a terminal FCF margin of 19.5%. All scenarios discount unlevered cash flows at the model-derived WACC of 11.29% and apply a Gordon growth terminal value.

SCENARIO	WEIGHT	REVENUE CAGR	TARGET FCF MARGIN	TARGET	UPSIDE
Bear	25%	18.0%	9.3%	$2.17	-71.4%
Base	50%	38.5%	14.3%	$5.18	-31.8%

Bull	25%	55.0%	19.5%	$10.32	+36.0%

Bear $2.17

The bear outcome hinges on working-capital drag and a slowdown in contract wins. If accounts receivable continue to rise and the $100 million+ Stark Future deal stalls, cash burn could intensify, limiting the ability to fund growth and compressing margins to the 9.3% target.

Base $5.18

The base path follows the current trajectory: revenue expands at 38.5% CAGR, gross margins stabilize near 28%, and adjusted EBITDA turns positive by FY2027. Continued execution of the Stark Future and European drone orders sustains the revenue base while working-capital improvements moderate cash outflows.

Bull $10.32

The bull case requires the full realization of the $100 million+ Stark Future multi-year agreement, additional defense contracts, and further margin expansion to 19.5% FCF. Under these conditions, revenue compounds at 55.0% and the company achieves a sustainable free-cash-flow profile that justifies a $10.32 valuation.

VALUATION FRAMEWORK

A live discount rate, not a static hurdle

The 11.29% WACC incorporates a market-derived cost of equity based on a beta of 1.80, a risk-free rate consistent with Treasury yields, and an equity risk premium that reflects the technology-intensive nature of the business. This discount rate is dynamic, adjusting as the company's beta decays toward a mature peer as scale and cash flow stability improve. Applying the 11.29% discount rate to scenario cash flows yields a probability-weighted target of $5.71 per share. The weighted average cost of capital therefore captures both the upside potential of a high-growth trajectory and the downside of cash-flow volatility inherent in early-stage battery commercialization.

FCF margin normalization is the engine

In FY2025 the company posted a free-cash-flow margin of -49.4%, reflecting heavy cash burn and a nascent revenue base of $71.9 million. The model assumes margin improvement as gross margins rise from 27% in FY2026 to a steady 28% in the base case, allowing operating leverage to lift the FCF margin to 14.3% by FY2030. The bear scenario moderates this improvement to 9.3%, while the bull scenario accelerates it to 19.5% as scale efficiencies and pricing power deepen. These margin trajectories are anchored in the observed gross-margin rebound from negative 21% in Q4 FY2025 to 27% in Q2 FY2026, and the management guidance of at least 28% gross margin for FY2026. The

KEY VALUATION PARAMETERS

Forecast window	**FY2026 to FY2030**
Live WACC	**11.29% (2026-10-20)**
Terminal growth	**3.5%**
Equity beta	**1.80**
Diluted shares	**143.5 million**

HOLD

over $5.71 blended target

model therefore treats margin expansion as a function of both volume growth and the capital-light operating model.

Mechanics are explicit

Revenue becomes EBIT, cash taxes come out, then we add back D&A, subtract CapEx and the change in net working capital to reach unlevered free cash flow, which we discount and divide by diluted shares.

Methodology. Detailed EBIT-bridge DCF on three scenarios, probability-weighted 25/50/25. Beta decays toward a mature peer over the forecast. Terminal value via Gordon growth. Targets are model-computed, not estimated.

BUSINESS & SEGMENT DRIVERS

Revenue prospects anchor on silicon-anode cell adoption and defense UAV demand

HOLD

Amprius's revenue is driven by two core levers. First, the adoption of silicon-anode cells in high-energy-density platforms creates a secular demand tailwind as drone manufacturers and electric-aircraft developers seek longer flight times. Second, defense procurement cycles, exemplified by the DIU contract and the $35 million follow-on UAS order, provide multi-year visibility and a premium pricing environment. The company's contract-manufacturing model allows rapid scaling of the SiCore platform without the need for heavy capex. This flexibility has already translated into a $24 million European drone order and a $100 million+ multi-year agreement with Stark Future, both of which are reflected in the FY2026 revenue guidance of at least $140 million. While revenue growth is robust, the balance sheet must support working-capital needs. H1 FY2026 operating cash flow was -$40.1 million, driven by a $16.8 million increase in accounts receivable. Managing this cash-flow timing will be critical to sustaining the growth trajectory.

What to monitor

- **Working-capital dynamics** – The rise in accounts receivable and inventory could pressure cash flow if order collections lag.
- **Contract execution risk** – The realization of the Stark Future multi-year deal and the European drone order will determine whether revenue guidance is met.
- **Margin trajectory** – Gross margin sustainability above 28% will be essential for achieving the targeted FCF margins.
- **Defense budget trends** – Changes in U.S. defense spending or procurement policies could affect the pipeline of UAV contracts.

+205.1%	**11.5%**	**-32.0%**	**-49.4%**	**$50.6M**	**-42.4%**

FY2025 REVENUE GROWTH	GROSS MARGIN	OPERATING MARGIN	FCF MARGIN	NET CASH	RETURN ON EQUITY

METRIC	FY2025 (AS REPORTED)
Revenue	$71.9M
Gross Profit	$8.3M
EBIT	$-23.0M
Net Income	$-44.0M
EBITDA	$-18.7M
Free Cash Flow	$-35.5M
Diluted EPS	$-0.35
ROE	-42.4%

RISK & MOAT MATRIX

The bear case

SHARE EROSION

Continued cash burn could force dilution if additional equity is raised, eroding existing shareholders' ownership and pressuring the stock price.

M&A DIGESTION

The company has not pursued significant acquisitions; any future M&A would require integration capital and could distract from core contract execution.

REGULATORY AND PRICING SCRUTINY

If defense regulators tighten pricing oversight for high-energy batteries, Amprius may face margin compression on existing contracts.

TECHNOLOGY DISRUPTION

Competing anode chemistries or breakthrough solid-state technologies could render silicon-anode advantages less compelling.

REINVESTMENT TREADMILL

Scaling the SiCore platform may require incremental capex and R&D spend, potentially re-introducing a capital-intensive cycle.

WACC AND DURATION SENSITIVITY

The 11.29% discount rate is sensitive to market volatility; a rise in the cost of capital would depress present values across all scenarios.

The strengths

OLIGOPOLY POSITION

Amprius occupies a niche in silicon-anode cells for aerospace, a market with few qualified suppliers, granting it pricing leverage.

FCF COMPOUNDING

The projected rise in free-cash-flow margins from -49.4% in FY2025 to 14.3% in the base case creates a compounding value driver as cash generation accelerates.

GEOGRAPHIC DURABILITY

Contracts span the United States and Europe, reducing reliance on any single jurisdiction and providing resilience against regional policy shifts.

PIPELINE OPTIONALITY

Beyond drones, the silicon-anode platform is applicable to electric vehicles and broader aviation, offering multiple growth avenues as the technology matures.

The bottom line on risk and reward

Amprius Technologies presents a compelling growth narrative anchored by a capital-light model, high-energy silicon-anode technology, and expanding defense contracts. The primary risk lies in working-capital execution and the need to sustain margin expansion. Our blended target of $5.71 per share, implying a -24.8% upside versus the $7.59 reference price, reflects a balanced view of upside potential and downside risk. Accordingly, we assign a Hold rating with valuation concerns, encouraging investors to monitor cash-flow dynamics and contract execution while recognizing the long-term upside embedded in the model.

Justin Wen, Senior Equity Research Analyst, Cascadia Analytics

FRANCHISE DEEP DIVE

Silicon-anode lithium-ion battery: high-energy-density cells for weight-critical platforms

Amprius's flagship franchise is its silicon-anode lithium-ion battery, marketed under the SiCore brand. The technology replaces conventional graphite anodes with silicon, delivering substantially higher specific energy, a critical advantage for drones, high-altitude platforms, and emerging electric-aircraft applications. The company's contract-manufacturing approach enables rapid scaling without the capital intensity of traditional battery fabs. Recent quarters illustrate the franchise's momentum. Q1 FY2026 revenue rose 153% YoY to $28.5 million, with gross margin flipping to 20% from a negative 21% a year earlier. Q2 FY2026 revenue doubled to $34.0 million, and gross margin climbed to 27%, the highest recorded to date. The company secured a $100 million+ multi-year agreement with Stark Future and a $24 million European drone order, both explicitly tied to SiCore cells.

- Revenue growth of 153% YoY in Q1 FY2026 and 126% YoY in Q2 FY2026 demonstrates rapid market adoption.
- Gross margin improvement from -21% in Q4 FY2025 to 27% in Q2 FY2026 reflects scaling efficiencies and pricing power.
- Adjusted EBITDA narrowed to a $1.0 million loss in Q2 FY2026, the closest to breakeven in the company's history.
- Major contract wins, including the Stark Future multi-year deal and European drone order, provide multi-year revenue visibility.

Our read: The silicon-anode franchise is transitioning from early-stage loss to a scalable, margin-positive engine. Continued contract execution and margin expansion will be the key catalysts that determine whether the company reaches the base or bull valuation outcomes.

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